

Summary of Findings: Visualization Task

Alt Mobility – Data Analyst Intern Assignment

Prepared by: Isha Dalvi

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Summary of Findings

Part 1: Key Insights from SQL Queries

1. Order and Sales Analysis

Majority of orders fall under delivered and shipped categories, indicating effective fulfillment processes.

Delivered orders generate the highest revenue, as expected, while pending orders contribute less and may indicate customer drop-offs or internal delays.

The average order value provides a baseline for revenue forecasting and helps identify potential upselling opportunities.

2. Customer Behaviour Analysis

A significant portion of customers are repeat buyers, showing a strong retention trend.

Several customers have a long history of transactions, indicating loyalty and product satisfaction.

Customers placing orders within shorter intervals could be high-value targets for subscription-based or loyalty programs.

3. Payment Status Analysis

Over 90% of payments are completed, but there are some failed payments.

Failed payments represent lost revenue and should be minimized through improved payment systems or customer support intervention.

4. Order Details Integration

Joining orders and payments tables revealed a small number of orders with missing or failed payment records.

This integration helps identify fulfillment vs. financial mismatches for further reconciliation or fraud detection.

Part 2: Customer Retention Analysis

What Was Done:

Conducted cohort analysis using customer first order month as the base.

Calculated retention rates as the percentage of customers who returned in subsequent months.

Visualized data using a retention heatmap.

Key Insights:

Retention is strongest in the first 1–2 months after the initial order, then gradually declines.

Some cohorts show surprisingly high retention in later months, possibly due to seasonality, campaigns, or onboarding quality.

Retention decay indicates opportunities for better engagement in months 3+ (e.g., reminder emails, discounts).

Visualization Description:

Rows = Cohort months (e.g., Jan 2023)

Columns = Months since first order (cohort index: 0 = same month, 1 = next, etc.)

Each cell = % of retained customers from that cohort

Visualized as a heatmap for quick trend detection.

Recommendations for Alt Mobility

- Target high-LTV customers identified through order frequency and revenue.
- Launch re-engagement campaigns after Month 2 to counter retention drop-off.
- Improve payment reliability to reduce transaction failure rates.
- Use cohort trends to evaluate onboarding changes and campaign effectiveness.
- Expand use of data dashboards to monitor KPIs like retention, revenue, and active users in real-time.

Summary of Key Performance Indicators (KPIs) & Visuals

Dashboard Title:

“Alt Mobility | Operational & Customer Insights Dashboard”

1. Monthly Active Customers

Objective:

To measure ongoing customer engagement and usage patterns.

Insight:

The trend shows a **steady growth** in monthly active customers over the last several months, indicating expanding reach or improved retention strategies. Occasional dips likely reflect seasonal effects or operational slowdowns.

Recommendation:

Focus marketing and support during months with lower activity to maintain consistent growth.

2. Monthly Revenue Trend

Objective:

To understand how monthly revenue fluctuates with customer activity and order volume.

Insight:

Revenue closely follows customer activity but shows **disproportionate increases** in some months, suggesting **higher average order values** or large enterprise orders. Identifying what contributed to those spikes can help replicate success.

Recommendation:

Analyze high-revenue months to identify drivers (e.g., campaigns, new services).

3. Order Status Distribution

Objective:

To evaluate order fulfillment efficiency and backlog.

Insight:

A **majority of orders are marked as “delivered”**, confirming operational efficiency. However, a non-trivial number remain “pending,” suggesting some issues in fulfillment or confirmation processes.

Recommendation:

Regularly audit pending orders and automate follow-up actions to reduce delays.

4. Payment Success Rate

Objective:

To assess reliability of the payment process and identify potential revenue leakages.

Insight:

The **payment success rate is high (typically over 90%)**, indicating a stable transaction pipeline. Still, even a small failure percentage could represent lost revenue.

Recommendation:

Set up alerts for monthly spikes in failed payments; review gateway logs or retry logic.

5. Top 10 Customers by Revenue

Objective:

To identify high-value customers for strategic focus.

Insight:

The top 10 customers contribute a **significant portion of total revenue**, showing the importance of retaining these users. Their activity can be used as a model for customer success strategies.

Recommendation:

Implement a **loyalty or premium service tier** for high-value customers to maximize LTV (lifetime value).

Final Note

These KPIs, combined with customer retention analysis, provide a comprehensive view of Alt Mobility's operational performance. The dashboard serves as a **decision-support tool** for marketing, customer success, and operations teams to **drive data-backed strategies**.